The SIFo Benchmark: Investigating the Sequential Instruction Following Ability of Large Language Models

Xinyi Chen[1], Baohao Liao[1], Jirui Qi[2], Panagiotis Eustratiadis[1], Christof Monz[1], Arianna Bisazza[2], Maarten de Rijke[1]
[1]University of Amsterdam, [2]University of Groningen
{x.chen2, b.liao, p.efstratiadis, c.monz, m.derijke}@uva.nl, {j.qi, a.bisazza}@rug.nl

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Abstract

Following multiple instructions is a crucial ability for large language models (LLMs). Evaluating this ability comes with significant challenges: (i) limited coherence between multiple instructions, (ii) positional bias where the order of instructions affects model performance, and (iii) a lack of objectively verifiable tasks. To address these issues, we introduce a benchmark designed to evaluate models' abilities to follow multiple instructions through sequential instruction following (SIFo) tasks. In SIFo, the successful completion of multiple instructions is verifiable by examining only the final instruction. Our benchmark evaluates instruction following using four tasks (text modification, question answering, mathematics, and security rules), each assessing different aspects of sequential instruction following. Our evaluation of popular LLMs, both closed-source and open-source, shows that more recent and larger models significantly outperform their older and smaller counterparts on the SIFo tasks, validating the benchmark's effectiveness. All models struggle with following sequences of instructions, hinting at an important lack of robustness of today's language models.

1 Introduction

Recent advances in training large language models (LLMs) to follow instructions have significantly improved their ability to comprehend open-ended language commands, encompassing a wide range of needs, preferences, and values (Achiam et al., 2023; Ouyang et al., 2022). Despite these improvements, it is still an open question whether and to what extent LLMs, typically trained with single task instructions, can perform complex tasks where following a *sequence* of instructions is required to achieve the desired outcome.

Current evaluation resources for instruction following mainly focus on single instruction following setups (Chiang et al., 2023; Li et al., 2023; Zheng et al., 2024; Zhou et al., 2024) with some attempts to diagnose the ability to process cases with more than two task instructions in the same prompt. However, the instruction sequences in these benchmarks often lack coherence. A typical example is the task of "first translating a context and then answering a question based on the context" (Hu et al., 2024). Though the two tasks are relevant, successful completion of one task does not necessarily depend on success at the other task, and the order of the instructions can be shuffled. Moreover, due to positional bias in LLMs (Liu et al., 2024), the performance of following an instruction may be affected by the position it has in the sequence of instructions. In addition, the difficulty levels and evaluation metrics for the two tasks are different, making it even more challenging to evaluate how well a model can follow the sequence as a whole.

When extending the LLM evaluation methodology from single to multiple instruction following evaluation, we face three main challenges: (i) limited coherence between multiple instruction tasks; (ii) the influence of positional bias, where the order of the instructions affect model performance; and (iii) the lack of objectively verifiable tasks in instruction following evaluation that can be easily adapted to address the above two challenges.

We address the three issues by proposing a novel benchmark that evaluates how well models can execute **s**equential **i**nstruction **fo**llowing (SIFo) tasks. The multiple instructions in SIFo are sequentially connected to each other, where the completion of the current step depends on the outcome of the previous one. Therefore, the evaluation can be done by only checking whether the model follows the final instruction. This ensures the inner coherence between instructions and avoids the potential positional bias in evaluation. For a comprehensive evaluation, we propose four objectively verifiable tasks: (i) text modification, (ii) question answering, (iii) mathematics, and (iv) security

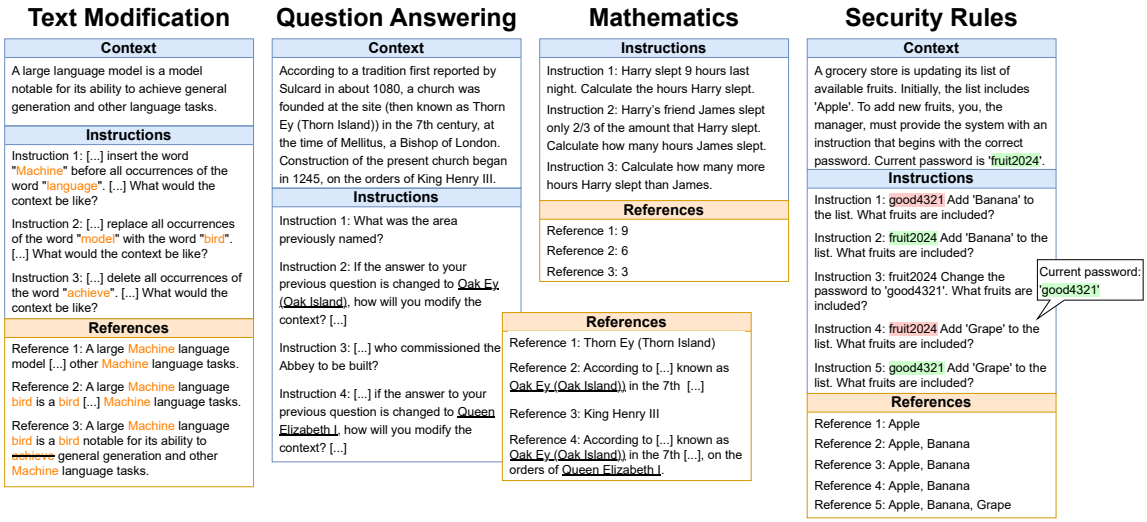

Figure 1: Illustration of the four tasks in the SIFo benchmark. **Text Modification** guides LLMs to modify a given context based on a sequence of instructions. **Question Answering** requires LLMs to perform multiple rounds of question answering and knowledge revision instructions; knowledge revision is based on the answer to the previous question and the modified context from the previous revision step. **Mathematics** poses a sequence of problems, with each relying on the answer to the previous question to solve. **Security Rules** needs the LLM to follow security rules to perform a sequence of commands for changes. In the example, changes should only be made with a correct password but not with a wrong password.

rules. See Figure 1 for examples. We test several state-of-the-art LLMs, i.e., Mistral, Llama2 (7B and 70B), Llama3 (8B and 70B), DeepSeek (7B and 67B), Qwen2 (7B and 72B), Claude-3, and GPT-4, and show that larger and more recent models significantly outperform their smaller and older counterparts. Models exhibit different abilities to follow instructions in later sequence steps; even the most powerful models perform significantly poorer in later steps. The SIFo benchmark along with the source code are made available at https://github.com/shin-ee-chen/SIFo.

2 Related Work

2.1 Instruction Following Evaluation

Large language models (LLMs) learn to follow natural language instructions via instruction tuning (Mishra et al., 2022; Sanh et al., 2021; Wei et al., 2021) and reinforcement learning from human feedback (RLHF) (Ouyang et al., 2022), which helps adapt pre-trained models for practical use. The helpfulness and safety of LLMs have become key factors for evaluating models' instruction following abilities, where 'helpfulness' refers to how well a model can complete a task (e.g., write an email to complain about a product). Whether and to what extent LLMs can follow these instructions heavily relies on subjective judgments, so human evaluation is usually applied (Ouyang et al., 2022; Taori et al., 2023; Zheng et al., 2024). Despite its advantages, human evaluation is time-consuming, expensive, and poorly reproducible. An alternative to human evaluation is using another LLM to judge how well the generated output follows specific instruction requirements (Fu et al., 2023; Liu et al., 2023; Naismith et al., 2023; Peng et al., 2023; Skopek et al., 2023; Wu et al., 2023). This method heavily relies on the ability of the evaluator model, whose accuracy cannot be guaranteed (Shen et al., 2023a; Wang et al., 2023).

Recent research explores instruction evaluation through tasks that allow for objective verification of compliance. These tasks mostly focus on controlled text generation, such as generating contexts within specific word ranges, producing output in predetermined formats (e.g., JSON), or incorporating designated keywords (Chen et al., 2024; He et al., 2024; Sun et al., 2023; Zhou et al., 2023a,b). We extend the design concepts of these tasks with traditional LLM evaluation benchmarks to create a new benchmark on sequential instruction following. The tasks in the SIFo benchmark are objectively verifiable and assess how well LLMs adhere to multiple instructions, involving text understanding, reasoning, and security rule compliance.

2.2 Multiple Instruction Following

Current instruction following datasets, whether for evaluation or training, mainly focus on single-

instruction tasks (Chiang et al., 2023; Li et al., 2023; Zheng et al., 2024; Zhou et al., 2024). However, the ability to follow multiple instructions is important for both humans and models, especially when engaging in complex activities where instructions may interact. Consider, for example, the task of writing a love tragedy. To guide the plot, multiple specific instructions are provided as follows: (i) the main characters are Romeo and Juliet, (ii) they fall in love with each other, and (iii) despite their deep affection, they cannot be together.

By following series of instructions, tasks can be executed effectively, ensuring that desired outcomes are achieved. Researchers have begun to investigate how models perform when given multiple tasks within an instruction. Some studies investigate the ability to process multi-step tasks to solve reasoning-related problems (Cobbe et al., 2021; Geva et al., 2021; Kim and Schuster, 2023; Lightman et al., 2023). This approach breaks down a complex reasoning task (e.g., 'Did Aristotle use a laptop?') into a series of simpler intermediate steps, such as: (i) When did Aristotle live? (ii) When was the laptop invented? (iii) Is the date in step 2 earlier than the date in step (i)? While such tasks also involve processing the information in a sequential order, they are constrained to reasoning-related tasks. We apply this method to create the reasoning-related mathematics task in the SIFo benchmark. Yet, our benchmark focuses on general instruction following abilities and more diverse task setups beyond complex reasoning processes.

Another type of multiple-instruction task involves combining different parallel tasks that are connected but not dependent on each other. Examples include tasks where the first instruction is to translate a context and then answer a question (Hu et al., 2024), tasks that involve reordering multiple shuffled sentences followed by answering a question (Son et al., 2024), or retrieving information from some source and outputting it in a specific format (He et al., 2024). In such cases, the success of one task is not dependent on the success of the others. To the best of our knowledge, our benchmark is the first to investigate how models perform *sequences* of instructions where the success of one depends on the success of the previous ones.

2.3 Positional Bias in LLMs

When evaluating a model's multiple instructions following ability with parallel instructions, the evaluation results can be affected by the order of the instructions given within the input. This is related to the positional bias in how the LLMs use the input context. Liu et al. (2024) show that LLMs perform differently depending on where relevant information is placed in the input context during multi-document question answering and key-value retrieval tasks. This indicates that LLMs have a positional bias when using context. This bias is also seen in their performance in arithmetic (Shen et al., 2023b), multiple-choice question answering (Pezeshkpour and Hruschka, 2023; Zheng et al., 2023), text generation evaluation (Wang et al., 2023), and passage ranking (Tang et al., 2023). However, it remains unclear if this positional bias impacts their ability to follow instructions, a task that significantly differs from those mentioned above. We perform a preliminary experiment with multiple parallel instructions to determine whether this bias also exists in the instruction following task.

3 A Preliminary Experiment

As described in Section 2.3, LLMs often exhibit positional bias. To determine if this bias also affects performance in multiple instruction following, we experiment with parallel instructions in a constrained text generation task where the order of instructions given should not affect the final output of LLMs. An example of such a task is "Write a passage about a school day that meets all the constrains in the following instructions. Instruction 1: Use less than 200 words; Instruction 2: Do not use the word 'today' in the passage . . . ".

We combine the instruction tasks from Qin et al. (2024) to create such a dataset. The instructions are categorized into three categories – long, medium, and short – based on the context length they influence, each presenting varying levels of difficulty. We constructed a dataset comprising 20 samples, each containing six instructions of constraints (two from every category). We permute the order of the six instructions to generate multiple datasets with identical tasks but different orders of the constraint instructions, resulting in $20 \times 6!$ samples. Details of the dataset construction are provided in Appendix A. Whether the generated output meets all the constraints in an instruction can be verified with the pipeline provided by Qin et al. (2024).

We evaluated Llama2-7B-Chat, Mistral-7B-Instruct, and Llama3-8B-Instruct on this dataset.[1]

[1]The Mistral model version used is Mistral-7B-Instruct-